

MIDTERM EXAMINATION

CS304- Object Oriented Programming (Session - 2)

Question No: 1 (Marks: 1) - Please choose one
Which part of an object exhibits its state?

- ▶ Data
- ▶ Operations
- ▶ Any public part
- ▶ Any private part

Question No: 2 (Marks: 1) - Please choose one
Inheritance is a way to

- ▶ organize data.
- ▶ pass arguments to objects of classes.
- ▶ add features to existing classes without rewriting them.
- ▶ improve data-hiding and encapsulation.

Question No: 3 (Marks: 1) - Please choose one
Suppose you have been given the following design,

"A person has a name, age, address and ----- . You are designing a class to represent a type of person called a patient. This kind of person may be given a diagnosis, have a spouse and may be alive".

Given that the person class has already been created, what of the following would be appropriate to include when you design the patient class?

- ▶ registration date and diagnosis

- ▶ age and -----
- ▶ ----- and diagnosis
- ▶ diagnosis and age

Question No: 4 (Marks: 1) - Please choose one

What problem(s) may occur when we copy objects without using deep copy constructor?

- ▶ Dangling pointer
- ▶ Memory Leakage
- ▶ All of the given
- ▶ System crash

Question No: 5 (Marks: 1) - Please choose one

this pointers are not accessible for static member functions.

- ▶ True
- ▶ False

Question No: 6 (Marks: 1) - Please choose one

A static member function cannot be declared.

- ▶ Static
- ▶ Implicit
- ▶ Explicit
- ▶ Virtual

Question No: 7 (Marks: 1) - Please choose one

_____ remain in memory even when all objects of a class have been destroyed.

- ▶ Static variables

- ▶ Instance variable
- ▶ Primitive variables
- ▶ None of given

Question No: 8 (Marks: 1) - Please choose one
Friend functions are _____ functions of a class.

- ▶ None of given
- ▶ object member
- ▶ non-member
- ▶ data member

Question No: 9 (Marks: 1) - Please choose one
_____, which means if A declares B as its friend it does NOT mean that A can access private data of B. It only means that B can access all data of A.

- ▶ Friendship is one way only
- ▶ Friendship is two way only
- ▶ NO Friendship between classes
- ▶ Any kind of friendship

Question No: 10 (Marks: 1) - Please choose one
The statement `objA=objB`; will cause a compiler error if the objects are of different classes.

- ▶ True
- ▶ False

Question No: 11 (Marks: 1) - Please choose one
Identify which of the following overloaded operator function's declaration is appropriate for the given call?

`Rational_number_1 + 2.325`

Where `Rational_number_1` is an object of user defined class `Rational_number`.

- ▶ Rational_number operator+(Rational_number & obj);
- ▶ Rational_number operator+(double& obj);
- ▶ Rational_number operator+(Rational_number &obj, double& num);
- ▶ operator+(double& obj);

Question No: 12 (Marks: 1) - Please choose one
Which operator can not be overloaded?

- ▶ The relation operator (>=)
- ▶ Assignment operator (=)
- ▶ Script operator ([])
- ▶ Conditional operator (? :)

Question No: 13 (Marks: 1) - Please choose one
To convert from a user-defined class to a basic type, you would most likely use

- ▶ a built-in conversion operator.
- ▶ a one-argument constructor.
- ▶ an overloaded = operator.
- ▶ a conversion operator that's a member of the class.

Question No: 14 (Marks: 1) - Please choose one
The technique in which we visualize our programming problems according to real life's problems is called

-
- ▶ structured programming
 - ▶ object oriented Programming
 - ▶ procedural programming
 - ▶ non of the given

Question No: 15 (Marks: 1) - Please choose one
In object orientated programming, a class of objects cans _____ properties from another class of objects

- ▶ Utilize
- ▶ Borrow
- ▶ Inherit
- ▶ Adopt

Question No: 16 (Marks: 1) - Please choose one
A C++ class is similar to -----

- ▶ Structure
- ▶ Header File
- ▶ Library File
- ▶ None of the given

Question No: 17 (Marks: 2)

Can we create an array of objects for a class having user defined constructor? Justify your answer.

Question No: 18 (Marks: 2)

Friend functions increase 'Programming bugs'. What is your opinion?

Question No: 19 (Marks: 2)

Explain two benefits of setter functions.

Question No: 20 (Marks: 3)

What are binary operators? Give an example of binary operators overloading using any class.

Question No: 21 (Marks: 3)

Give c++ code to overload assignment operator for string class.

Question No: 22 (Marks: 5)

Writ c++ code to overload subscript[] operator for String class.

Question No: 23 (Marks: 5)

Detect and correct compile time error(s) in the following code.

```
class Exam
{
char *ExamName;
int No_of_paper;
```

```
public:
Exam()
{
ExamName = "Final Term";
No_of_paper = 5;
}

void setname( char* name) const
{
ExamName = name;
}
void setpaper(int paper) const
{
No_of_paper = paper;
}
char* getname()
{
return ExamName;
}
int getpaper()
{
return No_of_paper;
}
};

int main()
{
const Exam exam1;

cout << " Exam = "<<exam1.getname()<<endl;
cout << " Numbe of paper = " << exam1.getpaper();

getch();
return 0;
}
```